The Hunt for Dark Energy with Jason Rhodes

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis The Nancy Grace Roman Space Telescope, scheduled for launch toward the L2 Lagrangian point, serves as a critical multi-purpose observatory designed to investigate dark energy through three distinct observational methods: Type Ia supernovae, galaxy clustering analysis, and weak gravitational lensing. By measuring cosmic acceleration over deep time from multiple angles, Roman aims to resolve the Hubble Tension—the discrepancy between the expansion rates of the early and late universe—potentially triggering a revolution in cosmological physics or confirming that dark energy is a constant. Beyond dark energy, the telescope will significantly advance exoplanet science by detecting "rogue" planets via microlensing and imaging close-orbiting exoplanets using an advanced coronagraph to search for biosignatures.

2. Key Arguments & Data

3. **Dark Energy Definition:** Current scientific consensus names it based on ignorance ("energy causing accelerating expansion") rather than identification, as we cannot directly see dark energy itself.
4. **Observational Location:** Roman operates at the L2 Lagrange Point (beyond the moon) to avoid atmospheric distortion and light pollution from thousands of low Earth orbit satellites (e.g., Starlink), which streak ground-based images.
5. **Mission Instruments:**
    - **Wide Field Instrument:** Spans 0.48 to 2.3 micrometers; features 8 distinct infrared filters to map billions of galaxies and supernovae.
    - **Coronagraph Instrument:** Creates an artificial eclipse to block starlight, allowing direct imaging of faint exoplanets using electron-multiplying CCDs capable of counting single photons.
6. **Three Methods for Dark Energy:**
    1. **Supernova Surveys:** Using Type Ia supernovae as "standard candles" to measure expansion rates over cosmic time.

2. **Galaxy Clustering:** Analyzing how galaxies cluster over cosmic time (gravity vs. dark energy push).
3. **Weak Gravitational Lensing:** Mapping invisible dark matter distribution via slight distortions in background galaxy shapes.

7. **The Hubble Tension:** A crisis where early universe measurements (e.g., CMB) do not match late-time measurements (expansion history); Roman aims to provide better late-time data to verify if the issue is new physics or imperfect measurement.
8. **Euclid Mission Update:** The ESA/NASA Euclid mission, flying for nearly three years prior to Roman's launch window, provides wide-field survey data. A key synergy was Euclid's pause to image the galactic center for microlensing observations to support Roman's high-resolution targets. If Euclid and Roman agree on results, confidence in cosmological models increases; if they disagree, it suggests systematic errors or observation bias rather than new physics.
9. **Exoplanet Demographics:** Microlensing can detect small Earth-sized planets and "rogue" (free-floating) planets that traditional transit methods miss.

10. **Biosignatures:** Roman's coronagraph will not produce detailed surface images (single pixel) but will allow spectral analysis (e.g., methane, oxygen) to search for signs of life on sun-like stars in our galaxy.

11. Actionable Takeaways

12. **Track Data Releases:** Follow up with the first major data release from the Euclid mission and wait for the Nancy Grace Roman Space Telescope's initial full data collection.
13. **Verify Cosmic Anomalies:** When reading headlines about cosmological crises (like the Hubble Tension), look for corroborating evidence from multiple independent techniques (supernovae vs. lensing) rather than assuming new physics without verification.
14. **Engage with Resources:** Visit the official NASA Nancy Grace Roman Space Telescope website for graphics and technical data, and follow Paul MacKerio's tour dates/podcasts for pop-culture educational outreach on these topics.

Full Transcript

Paul, we're on the trail of dark energy for this episode. Yeah. And we're hoping to find something. But if we don't find something, that means we found something. So everything's fine. And so I want to work out in the end. OK. Not exactly how science works, but we'll catch up on the latest of NASA's effort to decode this most mysterious thing in the universe coming right up. Welcome to Star Talk. Your place in the universe where science and pop culture collide. Star Talk begins right now. This is Star Talk. Neil deGrasse Tyson here, your personal astrophysicist. My co-host today, Paul MacKerio. How you doing, Paul? I'm great. Neil, good to see you again. All right, professional comedian. You got a Broadway show. Is it on Broadway or off Broadway? It was off Broadway and Broadway, but now we're touring around the country with it. We got to touring. We love that. Yeah. We love that. We don't want to keep you just for new... No, people want a lot of me, plus some...

My wife wants me out of the house, and the late show's over, so... Oh, that's right. That's right. You worked with the late show. I forgot about that. Yeah. Well, now it's... Everybody's on the street looking for gigs. Yeah, exactly. I'm selling lemon ice in an hour, so can we get this moving along? I got a cart that's not going to sell itself. So, subject of today is dark energy. Ooh. Is that what you are, Paul? You think? I am. Yes, I'd say dark, low energy. There's a telescope en route to its destination at one of the Lagrangian points called the Nancy Grace Roman Space Telescope, another generation of space telescopes taken the universe by storm.

And we have one of the world's experts on that, a friend and colleague, Jason Rhodes. Jason, welcome to Start Talk. Thanks for having me on. I'm super excited to talk about Roman and dark energy and all the science it's going to do. Yeah, very cool. Let me get your proper CV here. Senior research scientist at NASA's Jet Propulsion Lab. We all love JPL out there in Pasadena, California. It's one of the jewels in the crown of all the California gives the world. So thanks for being a part of that. And it's not your first rodeo with us. So thanks for returning. Thanks for having me back. I'm weird because Jason said he'd never come on again. He told me he

said, what's up with the I'm like, I don't know. I can't. So the envelope, the envelope of cash convert changed his mind. Is that what that was? Jason, the last time we had you on, you talked about the Euclid mission. We'll get an update on that later in this conversation. So so just tell us all about the Nancy Grace Roman telescope. First, who is Nancy Grace Roman?

And how can this telescope see something that is called dark? Like, what's up with that? Yeah. So Nancy Grace Roman was NASA's first chief astronomer back in the 1950s. And one of the things she did is she championed the idea of space telescopes. You know, before that, all the telescopes were on the ground. And she was, in fact, called the mother of the Hubble Space Telescope because that was NASA's first big flagship space telescope. And we decided to give her her due by naming NASA's next flagship space telescope, the Nancy Grace-Roman space telescope after her because of all the incredible work she did in convincing NASA and the world that putting telescopes above the atmosphere was a great way to study the universe. At the risk of saying something completely obvious, but I just want to make sure we're on the same page, remind people of the value of putting a telescope into space rather than on Earth's surface? Well, there's a few. The biggest one is being above the atmosphere. And so when we're kids, we learn this song, twinkle, twinkle, little star. And stars don't naturally twinkle. It's actually the atmosphere that's causing what we see as the twinkling. And so if you go above the atmosphere, you get a much clearer picture of stars and galaxies. The other big thing that allows a we're allowed to do when we go above the atmosphere is we can go to somewhere called the the Lagrange point Which is beyond the moon and it's dark there all the time So we don't have to only use the telescope at night So this is a more efficient place to have a telescope in terms of how often we can make observations? Well, we have a special explainer on Lagrangian points, which was fun to deliver. And I think it's actually done quite well on one of our YouTube channels. But that Lagrange point, that's not the only telescope at that Lagrange point. Isn't the James Webb Telescope?

at that same Lagrange point. That's right. The James Webb Space Telescope is there. The Euclid Space Telescope, which I talked to you about last time, is there. And there's telescopes there, like the Planck Space Telescope. This

is a telescope that the European Space Agency launched quite a while ago to study the earliest light in the universe. It's no longer active, but it was at L2, the Lagrange Point 2 as well. This is what this sounds very crowded. I'm just wondering. I was thinking the same thing. But one builds on the other, right? Hubble uses mostly visible light, James Webb's infrared light, and Roman is near infrared light with some visible light capability, right? So they're sort of a stepped process. Paul, you're showing off now. Paul, you're showing off. I'm done for the day. That's all the science you have to put back in the combo. I have a very small brain, and I just gave myself a headache.

So, Jason, I was out under the stars a few nights ago pointing out constellations for random passersby. And I couldn't help notice how many artificial satellites were crossing my field of view. Not 10 or 15 years ago, I would go the entire evening spotting maybe one or two of them at most three. And that evening I counted 40. Many of them are probably Starlink. from SpaceX, but what is that done to ground-based observing that may require that everything have to be put in space in the future? Can you just comment on that? Yes, so these multiple, you know, small satellites that are being put into low Earth orbit. Thousands. Don't say multiple. Thousands, thank you. Thousands. They create streaks in the image from a ground-based telescope.

But they also create streaks in the images from some telescopes that are also in low Earth orbit. So NASA's got a telescope called Sphere X, which is doing a survey of the whole sky in the mid-infrared. And that sees streaks from these thousands of telescopes. And that's another reason we want to send a telescope like Roman. out to the Lagrange point, because then we're very far away from Earth. We're also very far away from these thousands of low Earth orbit satellites. So we can add that to the list of reasons we're sending Roman to the Lagrange point. So you agree it's pollution. It's another form of pollution in a sense. That you have to work around and factor in, right? Yeah. In terms of developing this technology. You could say it's a data processing challenge. How's that?

I'm Brian Futterman, and I support Star Talk on Patreon. This is Star Talk with Neil deGrasse Tyson. What is the Nancy Grace Roman Telescope

going to be looking at that enables it to track dark energy? Because last I checked, you can't see dark energy. So what's up with that? The Nancy Grace Roman Telescope is going to have three different ways of measuring dark energy. And dark energy is the name we give to whatever is causing the accelerating expansion of the universe.

And we don't know what that is. And we don't know what that is. It's really the name we give to our ignorance of what that is. That's why we call it dark. Well, I think dark is a little more elegant than we have no effing idea energy. You know what I mean? It just feels a little more elegant on some level. A little more mysterious. OK. Yeah. So we have to study this dark energy very indirectly because we can't see it. And so one of the ways that we study it is with supernova. And a supernova explodes, and it's what we call a standard candle. So we know how bright it was and how far away it was. And by looking at those over cosmic time, we can see how the universe has expanded over cosmic time. There's another way that we're going to use with Roman. And let's look at the clustering of galaxies. Yeah. So the expansion rate over time, the extent to which that doesn't match what matter would have it do, you didn't ascribe the rest of that to the influence of dark energy. Is that a fair way to say what you just described there? That's exactly right. In a universe with just matter and gravity, we would expect the expansion to be slowing down due to gravity. In the 1990s, two teams simultaneously were trying to find out how much the universe was slowing down due to gravity, and they were using supernova, and they both came up with the same answer, which was it wasn't slowing down, that expansion was speeding up. And it was great that there were two teams because both teams thought this can't be right. And in terms of what they thought was happening, they were correct. There was more to the universe than just matter and gravity. There was the dark energy. And one of the great lessons there, of course, is a singular scientific result, especially if it's either unexpected or weird, it's not really a scientific result until it's verified. So that's great. I wonder what would have happened if it was just one group that came up with that result.

That would have been a little weird, right? That's exactly right. And that's why we're using multiple techniques with Roman to study this dark energy. And in fact, there's multiple telescopes out there with a primary goal of

studying dark energy because we want more than one telescope to understand this really mysterious aspect of the universe. And so we've all heard about this Hubble tension that came about from the observation of... of galaxies, misbehaved galaxies in the early universe with the James Webb telescope. Would the Nancy Grace Roman help address this Hubble tension, this cosmology and crisis that headlines are talking about? Absolutely. Roman is coming at just the right time to address this Hubble tension. And this tension arises from us having very good measurements of the early universe.

and thinking we understand the physics that gets us from the early universe to the universe we see today, but the universe we see today not quite matching what we would expect given that early universe and the physics. So something we think might be wrong. It might be our understanding of the early universe, but we've got really good measurements of that. It might be we don't know the physics that gets us to the later time universe that we see now, or maybe it's imperfect measurements of the late time universe. And what Roman is going to do is it's going to give us really, really excellent measurements of the late time universe. And so while scientists were supposed to be sort of agnostic about what we're studying, most of my colleagues are hopeful that the answer is new physics because that's what's exciting. You know, what's exciting for a scientist is when they look at something, they go, huh, I didn't see that coming. As you feel confident that that's 100 times the span that the Hubble telescope can cover, right? So you're all feeling confident that some answers will come regarding dark energy, but what would surprise you more? What dark energy is? Or maybe that our understanding of gravity or physics is incomplete. We have to come up with a new science. Do you all have a gut as to what you think you might be getting through, Roman? Well, I really hope it's this new physics.

And I don't know exactly what that new physics might be. And the reason I hope that is all scientists hope to live through a revolution in the understanding of their field. That's the most exciting time to be a scientist. And I was in graduate school in the 1990s when we had this revolution of Learning that there was dark energy, and so I'm really fortunate to now be you know in the midst of my career working on something that might have another Revolution which tells us what dark energy is. You know, I just hit

on something. I just hit on something real quick. You scientists are brilliant because even if something fails you'll say it didn't fail.

We just found something incredibly informative that we didn't know. I want to be you. You guys have figured this out, man. Because Neil, although we say a failure isn't a failure in science, it's just opening up another avenue that we haven't thought of, right? But so Jason, wouldn't changing dark energy, because in general relativity, dark energy is just a constant term in that equation. So would a changing dark energy term that is dark energy is a function of time or size of the universe that kind of means new physics right because you're not getting that for free out of general relativity because general relativity requires that it just be a constant number or you still think of yet some other physics nobody has even conjured yet no certainly some of the models of dark energy have evolving dark energy, a changing dark energy. Those aren't the simplest models, which is the constant dark energy or so-called cosmological constant. But what we're saying is tantalizing data from other telescopes that tell us that maybe that simplest model isn't the right model, and we do have perhaps an evolving dark energy. We have an incomplete rather than wrong, incomplete understanding of how gravity works. Of course. See how we do that, Paul? You got that pole. Wow. Hang on. I got to make sure I still have my wallet on me. You guys are slippery. You are a slippery group of people. So Roman's like doing...

Roman's like doing the job of the USS Enterprise, right? It's exploring the universe. The difference is when Roman finds a new planet, nobody immediately falls in love with a green alien, right? I mean, is that the way the difference here? We hope not. So is time-dependent dark energy what you're calling new physics? Or is there some other physics undreamt of that you're considering in this scenario? There are many many different theorists with many many different ideas for what this might be and Time time evolving dark energy is one of them, but as I said Modified gravity that that's what we call the fact that Einstein's theory may have been incomplete You could have modifications to our theory of gravity that would explain some of the observations as well And one of the reasons we're going to study this phenomenon with multiple different ways with Roman is we want to be able to distinguish between time-evolving dark energy and a modification to gravity and all these other theories. And any one way of

probing dark energy might not get at it, but multiple ways will allow us to study the whole gamut of theories. So Jason, like the James Webb Space Telescope, which was tune for the infrared and conceive, because I remember when we were specking that telescope in the Decadal Survey now three decades ago, we wanted it to be able to specially get access to the early universe where galaxies were being born. It just so happens that same design telescope can probe nearby gas clouds and observe stars and planets being born.

which is a fun sort of duality of that telescope. You're telling me that you're going to see supernova across the universe, across time. Does that capacity give you other utility in questions we might have about the universe that might have nothing to do with dark energy? Oh, certainly. Supernova survey is only going to look at a small area of the sky and wait for these supernova to happen. But the other surveys that we're going to do to study dark energy are going to look at big swaths of the sky. And one of the ways we do that is the camera, the wide field instrument in Roman, is about 100 times as big as the comparable camera in Hubble. So we'll look at hundreds or thousands of times more area of the sky. And we'll look at things that allow us to study dark energy but we'll put that data in an archive and that archive data is going to be useful for a huge range of astrophysical questions literally for decades to come from planets in our solar system to planets throughout our galaxy to studying galaxies throughout the universe.

There's going to be a huge range of science questions that we can address with this same data that we're taking to study dark energy. I just have a quick question. Can we talk about the telescope for one moment so that you talk about multiple ways that it's going to do this? Neil's going to yell at me for knowing this, but it's a total spectrum range spans wavelengths from 0.48 to 2.3 micrometers. The telescope can break down light coverage through two core instruments, which is a wide field instrument and a coronagraph instrument. Those are the two main vehicles through which this information will get taken. So the coronagraph is a whole different instrument that's not useful for studying dark energy. So what a coronagraph is, is it's an instrument that allows us to take direct images of exoplanets, planets outside of our solar system and what it does is it creates an artificial eclipse and it eclipses the star so that we can see the much fainter planet

that's orbiting the star and if you think of if you've ever seen a solar eclipse it's it's an amazing thing to see the moon eclipses the sun and you can see the light from around the Sun. That's called the corona, you know, Latin for crown around the Sun. That's the only way you can see that light directly around the Sun because you have to block out the very bright Sun. So the Roman coronagraph is a technology demonstration instrument that uses a bunch of new technologies to give us about a hundred to a thousand times better ability to block starlight and see very faint planets orbiting those stars.

Can I ask sort of a can I ask a lay person's question for a moment? It seems to me Rome is counterintuitive in a way if we want to really try to drill down on dark energy Does it exist? What is it a wider swath? It seems to me that a more narrow focused Telescope analysis would get us to that answer more so than a wide Angle I know that sounds probably rudimentary, but he said he's doing both He's got the narrow field of view that'll look look and wait for supernovae because if you've taken enough of the universe you're pretty much guaranteed to get a supernova going off practically on schedule so he's got that and that's the sort of the pencil beam view through the greatest depths of the universe and that's one way and then another way so are you looking for in your wide field view what are you looking for that could reveal the presence of dark energy.

In that field of view so in the very wide surveys we're gonna have two different ways of studying dark energy one is looking at the clustering of galaxies and how that clustering of galaxies. Changes over cosmic time and how close galaxies are to each other is sort of an interplay between gravity pulling them together and the dark energy pushing them apart and so to do that. We need a huge statistical sample of galaxies, and that's why we want to look very, very wide. But you have to disentangle the effects of dark matter on top of that. That's right. That's the other big component of the universe. But how do you disentangle something that you don't even know exists? That was my next question, Paul. It was not. Don't lie. We don't know what dark energy is. We don't know what dark matter is. And there they are.

intermingled in the clustering of these galaxies. And what hope do you have of disentangling those two effects? Roman is going to be amazing at mapping where this invisible dark matter is. And how we do that with Roman, or any telescope, but Roman is going to be just amazing at this, is we look at a distant galaxy. And the light from that distant galaxy comes to us and the path of the light is bent by that intervening dark matter. that we can't see but we know that dark matter is there because it has gravity and gravity bends space and the light travels through space so those distant galaxies will have slightly deformed and warped shapes and by looking for coherent distortions in galaxy shapes across many thousands or millions or in the case of Roman hundreds of millions of galaxies out there in the universe we're going to be able to create very intricate maps of where that dark matter is and that's the third technique we're going to use to study dark energy by mapping the dark matter using this technique called weak gravitational lensing and the gravitational lensing is the bending of light and the weak just means that We're seeing very small distortions in the shapes of these background galaxies. If you map in detail the dark matter and we can trust what those maps are and the distribution is, then any effects beyond that you get to credit or blame dark energy. Is that a fair way to characterize that exercise? That's exactly right. If we believe that our knowledge of gravity is 100% complete and accurate. So this becomes the problem. And this is why I want my taxpayer dollars back. No. But this is why you think three techniques. Right. And so, I mean, look, we're going to map billions of galaxies. Is there a more efficient way to do this? I mean, if men in black taught us anything, it's like go to a locker in Grand Central Station and it's there, right? It's C18.

that's the place to find your galaxies right there, right? Yeah, that's a miss the memo on that one. Yeah, we're still doing it the old fashioned way. Thank you. Yeah, we still use telescopes. Yeah. Let me get back to the chronograph for a minute. So with the chronograph, you're just piggybacking on a telescope that had other priorities. What did you find some extra sort of square footage on its outer surface and slap on and the whole other? I mean, what was the The planning for this, did you just have exoplanet people looking outside over the fence? Can you let us in? Can we join this launch? Yeah, I mean, how did exoplanets show up in the portfolio

here? Well, first I'll say that the wide field instrument is also going to do a survey. Maybe we'll talk about that later to find a bunch of exoplanets.

But when you talk about, that was always a plan for Roman since the very beginning. But when you talk about how the coronagraph came to be part of this telescope, the original plan for Roman was a smaller telescope than what we have now, about a meter and a half across instead of the 2.4 meters we have now. But when it came time to start building Roman, there was another government agency that had a surplus telescope already built that said to NASA, hey, do you have a use for this? And NASA put a group of us on Roman, you know, studying this. And they said, hey, would it be better to have a bigger telescope? And I'll tell you the answer is pretty much always yes, it's better to have a bigger telescope. But when we got this new bigger telescope, we said, you know what, we could add a coronagraph to this telescope because it's now big enough to add a coronagraph.

that tests all these new technologies. So the coronagraph itself is not considered a science instrument in the traditional sense for NASA because its goals are really to test a bunch of new technologies for doing these exoplanet observations. But the best way to test these technologies is to do science with the instrument. So that's what we're going to do. We're going to do a bunch of science with this coronagraph. So listen, I have a question. So most planet finding methods discover planets by looking at the star, but talk about microlensing here, which if it finds a planet, it focuses on what they do to the light of a completely different star. Is that why Romans expected to uncover planetary systems we've been sort of blind to up to this point? Absolutely. So over the past 30 years, we have discovered about 6,000 exoplanets and most of them have been discovered by looking at the star and seeing the star blink or seeing the star wobble. They're indirect ways of detecting planets so you don't actually see the planet. So one of those techniques that hasn't been used very much is called microlensing and in microlensing you wait for two stars to sort of pass in front of each other and the foreground star amplifies or lenses the light from the background star. So if you're looking at the light of that background star, you see a blip. So if I hold my finger up, which I'm doing right now, my left index finger, and I move my right index finger across, at some point they cross, and that's that

little blip that you talk about in my field of vision, right? That's right. That's right. And if that foreground star has a planet, you see a secondary blip.

secondary smaller blip tells you you've discovered a planet and so this technique allows us to find planets and what's really exciting about it is most of the other techniques they allow us to find really big planets or planets that are that are very close to their star. Whereas microlensing will let us find smaller planets like planets the size of earth and planets further from their stars. So it's going to allow us what we call complete a demographic census of solar systems. We're finding the big planets and the close planets and the far planets and the small planets. So we can see really all of the planets that we expect to see. And then the politicians will get involved and there'll be a whole redistricting of the entire universe. And there'll be manipulation of the thing. So let me understand this, guys. Think about this. We will most likely discover We've already 6,000 exoplanets. So somewhere, think about this, there's an alien Neil deGrasse Tyson with a show called Earth Talk. And they are literally still answering a question that was asked five episodes ago. Neil, their version of Neil is still going on and on and on. It's going to be, it's going to be, it's the ying and the yang of life right there. So Jason, it seems to me that it's microlensing. You could also detect planets.

that are homeless that are just free floating in space in principle. You should be able to detect those as well. Is that right? That's right. And in fact, we call those planets rogue planets. These are planets that are not bound to a star. And the microlensing is really the only way we can detect them from those small blips in the background star. And so the way that we do that with Roman is you're going to stare at the center of our galaxy and just keep taking images over and over and over, and the center of our galaxy is very rich with stars. And you're going to do that and look for these telltale blips, and some of the planets we find are going to be these rogue planets that are not bound to a star. And we think that there might be more of these rogue planets than there are normal planets, what we consider normal planets, planets around a star in our galaxy.

And so they're a really interesting discovery that we're sure we're going to make with Roman. Jason, you and I came through graduate school slightly,

I think it was slightly ahead of you, maybe by 10 years or so. While I was in graduate school, there was the discovery, primarily from computers being applied to this problem, that the solar system or any star system might have formed with multiples of the number of planets that ultimately survived, right? So we have eight planets, get over it, eight planets. Oh, here we go. Poodlehuggers get over it. Right. Right. On the Earth Talk show, there's this alien Neil Tyson going, I never heard of Pluto. So we could have started with 30 planets, but only some fraction of those.

end up in stable orbits, and the rest fall into the Sun or to Jupiter get jettisoned. So it was the first awareness that there might be more planets wandering space than in orbit around Earth. And I presume that's just a given today. Is that correct? Yes, because we have detected some of those. And based on, you know, when you detect something, you can calculate how likely was it for us to detect this. And we've detected enough of them that we think they're very, very common. out there in the universe, but we haven't really detected enough to know their demographics. And that's what Roman is going to do. And understanding how many of them are and their properties is going to tell us a lot more about how planetary systems form. Here's yet another fact from Lost in Space, the latest collaboration between StarTalk and National Geographic.

1995, we discover the first authentic exoplanet, a planet orbiting another living star. So, anyone born from 1995 onward? I now declare each one of you to be Generation X, a planet. If you like that fact, you can find 4,999 more in Lost in Space. 5,000 facts to help navigate the universe. Lost in Space is now available for pre-order, wherever books are sold. justifiable confidence that you have in microlensing and the ability to detect these blips. But what if these blips aren't planets? It's just some random dark energy stuff that we haven't figured out yet. How do you, because Neil always talks about sort of having to do your experiment and then check it again and check it again and have various levels of evidence. How do you then confirm that that blip is truly a planet?

Or it's not some stuff that we don't know about yet, or don't understand. We think that stars form planetary systems, and that's what we're looking for. But one of the things that's interesting about these microlensing planet

detections is we're waiting for the chance superposition of two stars. And when we see that blip that's telling us we think that there is a planet. But we're never going to get that chance superposition of those stars again because those stars are all moving So we have no way of following up these planets that we find through microlensing We don't have a way of confirming that it was indeed a planet or even seeing that planet or or evidence of that planet again So it's really like the transiting method where you get to repeat the the planets transit across the surface of the star and get an orbital period. And so that gives us confidence that it's something orbiting a star, probably a planet, right? But you have no orbital information in these data. Exactly. But you might require future cosmic microwave background experiments, gravitational wave observatories, other surveys, right? So you're not even sure yet, and I'm not making this as a criticism, If we don't know what we don't know, we don't even know what we need to create to figure out what we don't know, right? Isn't this the challenge here? Well, that's always the challenge. And that's why there's been this centuries-long progress in building better and better telescopes. And often what happens when we build a better and better telescope is we answer the questions that we had that were the reason we built that telescope, but we end up with more and more questions and more interesting questions. We said, well, we're going to have to build another telescope to study this new thing that we found. And one of the fields where that is becoming more and more apparent is exoplanets. Because if you look at Hubble Space Telescope, that wasn't really built and designed to study exoplanets, but it's done amazing work.

in exoplanets. And Neil, you mentioned your work on the Decadal Survey, developing the James Webb Space Telescope. And I don't think when that was happening, understanding exoplanets was a big part of what people were talking about. But some of the most amazing discoveries of the James Webb Space Telescope have been discoveries about exoplanets. And now Roman is a telescope where one of the prime drivers along with dark energy is discovering exoplanets. And if we're talking about how can we study an exoplanet again after we've found it, that's where this coronagraph instrument comes in. Because then we can look directly at the planet. We're not looking at the star and waiting for the star to blink. We can look at the planet, we can take an image, or we can take a spectrum.

of the planet and we're developing technologies and testing out those technologies on this Roman coronagraph that will allow us in the future to build a better, more powerful coronagraph on a future telescope that is going to use those technologies to look at planets and look for what we call biomarkers. Those are signs of life. So we are in the stages of developing the technology that's going to allow us to look for signs of life on Earth-like exoplanets around sun-like stars in the neighborhood of our galaxy close to us. But Jason, if you're going to take pictures of planets, what we really want to see is like oceans and continents, and then you zoom in and see dinosaurs. So how many pixels big are these images?

Is it more than one pixel of occupied by the image of the planet that you've successfully shielded from the glare of the host star with your coronagraph? How much of an image of the planet is that? Yeah, I wonder if we've done ourselves a disservice because sometimes when people have exoplanet discoveries, they have artist's recreation of the exoplanet and you see these these things and what we're talking about are pretty much single pixels. But in that pixel, if you spread out the light, there's an incredible amount of information. And you can, for instance, if you looked at that pixel, that planet as a single pixel, if you were able to look at Earth from tens of parsecs away, you would see differences as the seasons change.

in what the Earth looks like. And so we can look for things like that. Even though it's one pixel, there's an incredible amount of information if we get a spectrum and spread out the light in that single pixel. And just to restate what you just said, but slightly differently, all the light from Earth, just to remind people what a pixel is doing to your image, if the planet is smaller than the pixel resolution, of your camera, then all of the light is blended into one signal through that pixel. Fine. And now you say, this is what it looks like in June. What does it look like in December? And if that changes, that gives you some information, even though it's not detailed information about what's going on, correct? That's right. And if you were to take a spectrum of the Earth, You would notice some interesting things in that spectrum. You would notice, you know, methane and oxygen. And these are, we think, probably signs of life. And this is where infrared comes in. Can we talk about that for a minute in terms of this particular telescope? Yeah, it's from, you know, the near ultraviolet.

through the optical, you know, what we can see with our eyes and then through the near infrared. So not as far into the infrared as like the James Webb Space Telescope. And there's two different, the two different instruments have two different types of detectors. And so the wide field instrument has the detectors that go through that full range that you mentioned, Paul. allows us to see optical, you know, visible light and the infrared light. And that infrared light uses it uses eight distinct filters, right? To peer through. Yeah, I don't remember. I don't remember the number. You may know the. Yes, it's eight. Do I have to do your job or my job? I should. I am not getting paid enough for this. This shirt. Yeah. I owe you this shirt. Jason, he's earned a free shirt, not at the commissary.

And so that uses and these are very advanced detectors that have been developed over you know many decades and they're the same family of detectors that that are on for instance the Euclid space telescope that we've talked about you know previously so these are detectors that NASA has honed to be able to do this astrophysics the coronagraph instrument is only uses really visible light or close to visible light. And so it doesn't have that full wavelength range. And it uses CCDs. And these are very common type of detectors. But it uses CCDs that are a new type. And that's one of the new technologies that I was talking about. And these CCDs in the coronagraph are called electron multiplying CCDs.

And what that means is they can count single photons as they arrive. And that's really important because when we're doing the coronagraphic observations, we're having to block something like 100 million or a billion photons from the star for every single photon we get from the planet. And so every single photon we get from that faint planet is incredibly precious and rare and important for our understanding. And so having these special CCDs that can count one photon at a time is one of our new technologies that we're developing for the Roman coronagraph. Almost perfect efficiency on that level. That's right, yes. It's blocking out intense starlight, right, to directly photograph these. Coronagraph instrument blocks it out so that as much as possible so that you're only getting the planet light fall onto that CCD detector. By the way, when the coronagraph was invented, I think more than a century ago, it was a brilliant device because you can simulate a total solar eclipse in broad... We use that on our own sun. We have a

coronagraph instruments just to pretend like there's an eclipse so that you can see what's going on around it. So it's a brilliant bit of optical engineering. So Jason, we got to land this plane. Just remind us what Euclid was doing and give us an update on Euclid before we...

before we take us out. So Euclid is a European Space Agency telescope with NASA involvement to study dark energy. The Europeans don't even get me started with those people. They're fantastic partners. You're good. Especially fantastic. Now that American scientists are looking for jobs. All right. We're getting the phone call from Europe. Right. OK. Go on, Jason. And so Euclid is doing this survey of the sky. It's going to survey an even wider area than Roman will, but not quite as deep as Roman does. So they're very complementary surveys. And Euclid has been flying for almost three years now. And we'll have our first big data release forthcoming. And so I'm excited about that and the cosmology science that that's going to enable. But one of the things Euclid did is it stopped its big wide cosmology survey for a day last year, and it looked at the center of our galaxy for that day as a precursor to the Roman images that are going to do microlensing. And when we talked about the microlensing observations, I mentioned that you're waiting for the stars to move and sort of eclipse each other, come in front of each other, and by taking those really high-resolution Euclid images years before the Roman images will become available in that field, we're extending the time baseline over which we can see those stars in the center of our galaxy move. So it's a really exciting way that Roman and Euclid are able to work together to do even more science than either one would on their own.

So telescopes scratching each other's back, telescopes scratching each other's back? Absolutely, yeah. But if they both tell, so if they both tell the same story about dark energy, then confidence in our cosmological model grows, right? But if they don't, and there's a discrepancy, which one do you trust? Which, where do you place more of your confidence? The American one, the American one. Yeah. Wait, wait, just Paul. Paul. One could be wrong. The other could be wrong or they both could be wrong. Right? I mean, you have to be open to all possible variants. But even the use of wrong is subjective. I'm not suggesting somebody's right or somebody's wrong. They just don't match up, right? So then, which do you, I mean,

because you always say there's not right or wrong really in science, right? So like, is there some, is that stronger evidence if they don't match up that there's something fundamental about the universe that current theories are missing?

No, in fact, it's the opposite. If they don't match up, it's probably telling us that our observations have problems rather than our understanding of the universe. Because the answers from the two telescopes... You mean the techniques, our techniques of observation. Well, it's either the techniques or there's all sorts of... Mathematics. Systematic effects is what we call them. Are we, is the telescope shaking a little bit? Do we really understand what's happening with the detectors and the camera? So there's lots of little things, and that's why it's important for us to cross-compare these. And one of the things that's exciting about these two telescopes is Euclid is going to take, as I said, a very wide view. So it's going to have a large number of statistics, more galaxies and things like that than Roman.

But Roman's going to take a deeper view and with multiple passes. So it's going to have internal cross checks. And so if we want to compare the two, one is going to get a better statistical sample and one is going to get a higher quality sample of the universe. And so it's really two different ways of doing this. And we want to keep doing it until their answers match with each other. We understand why they don't match. Yeah, and science, I mean, this is the great challenge of doing good science is how do you trust your data? How do you trust your instruments? How do you trust anything that's coming back? And what kind of internal checks and balances do you have to decide that? And where do you, and you got to leave your bias at the door because if you have a bias that likes one result that hasn't been checked, you're not as good at analyzing the integrity of that result if you can be influenced by the bias you bring into the room with you. So, Jason, this has been a delight to catch us up on this. Maybe we can get you back on the first data release of the Grace Roman Telescope. Yeah, then you can explain how you didn't make any mistakes and you're just discovering new stuff. Whatever. Love to. All right. Jason, once again, thanks. Oh, wait, would you say dark matter?

is a new understanding of gravity or just a new particle? Dark matter? Dark matter, I think all the evidence points to a new particle. Okay, and dark energy? What the hell is that? I wouldn't bet, yeah. I wouldn't even bet, yeah. Okay, there you go. I don't know enough to bet, yes. And so does the Grayser Roman Telescope have its own website on NASA? Oh, absolutely. If you Google Nancy Grace Roman Telescope, you'll find, as is typical for NASA, you'll find a website with all sorts of information, really cool graphics and explanations for all the science it's going to do. And NASA has been extraordinary in all their missions about public dissemination of what the mission is about, the cost, the benefits, what we're going to learn. And so this is just another example of that. So delighted to learn this. And Paul, where do we find you?

When are you next in my town? I am going to be Trump touring people go to Paul McCurio.com see all my tour dates and my podcast Inside Out with Paul McCurio wherever you get podcasts. Yeah, McCurio MEC. You are IO and follow me on social media. So I get more of a following and I can. All right, that's all the time we have. I'm delighted for this update because everyone cares about dark energy. I guess you're right. dark energy. This has been another installment of Star Talk. Neil deGrasse Tyson here bidding you to keep looking up.